

# Ironwood, the core

how Zcash verifies a shielded transfer without seeing it

m@rek.onl

## What a validator checks

1. the coin **exists** — look it up
2. the spend is **authorised** — script and signature
3. it is not spent **twice** — delete it
4. value is **conserved** —  $\sum \text{in} \geq \sum \text{out}$

Every check reads the transaction in the clear: which coin, whose key, how much.

## Privacy, precisely

Hide the sender, the receiver and the amount — while a stranger still runs all four checks.

| check      | mechanism                                                          |
|------------|--------------------------------------------------------------------|
| exists     | a commitment in a Merkle tree, membership proved in zero knowledge |
| not twice  | a nullifier: a keyed serial number, revealed once                  |
| conserved  | homomorphic value commitments and a binding signature              |
| authorised | a proof of key knowledge and a re-randomised signature             |

A zero-knowledge statement per Action carries part of each; one proof per bundle.

# Anchors

## Arithmetic with a modulus

For a prime  $p$ , the field  $\mathbb{F}_p$  is arithmetic modulo  $p$ .

$$\mathbb{F}_{97} = \{0, 1, \dots, 96\}, \quad 96 + 2 = 1, \quad 5 \cdot 39 = 1.$$

Every nonzero element has an inverse:  $5^{-1} = 39$  in  $\mathbb{F}_{97}$ .

Addition, subtraction, multiplication, division by nonzero elements. The protocol's fields have  $p \approx 2^{254}$ ; the toy  $\mathbb{F}_{97}$  returns in the proof.

# A group and a hard problem

$$\mathcal{E}(\mathbb{F}_p) : \quad y^2 = x^3 + 5 \quad \text{over } \mathbb{F}_p, \quad p \approx 2^{254}, \quad |\mathcal{E}| = q \text{ prime}$$
$$[a] G := \underbrace{G + G + \cdots + G}_{a \text{ times}} \quad \text{given } G \text{ and } [a] G, \text{ find } a : \text{ infeasible}$$

$G, [a] G \in \mathcal{E}$  points;  $a \in \mathbb{F}_q$  a scalar; both fields  $\approx 2^{254}$ .

- Pallas: points over  $\mathbb{F}_p$ , scalars in  $\mathbb{F}_q$ .
- Vesta: the same curve with  $p$  and  $q$  swapped — a cycle.
- Everything below is built from  $[a] G$  and the discrete log.

# Commitments

$$C(v, r) := [v] V + [r] R \quad V, R \text{ independent points}$$

$$v, r \in \mathbb{F}_q; V, R, C \in \mathcal{E}.$$

- **Hiding:**  $r \mapsto [r] R$  permutes the group, so  $[v] V + [r] R$  has the same distribution for every  $v$ .
- **Binding:** two openings of one  $C$  give  $[v_0 - v_1] V = [r_1 - r_0] R$ , hence the forbidden  $\lambda = (v_0 - v_1)(r_1 - r_0)^{-1}$  with  $R = [\lambda] V$ .
- **Homomorphic:**  $C(v_1, r_1) + C(v_2, r_2) = C(v_1 + v_2, r_1 + r_2)$ .

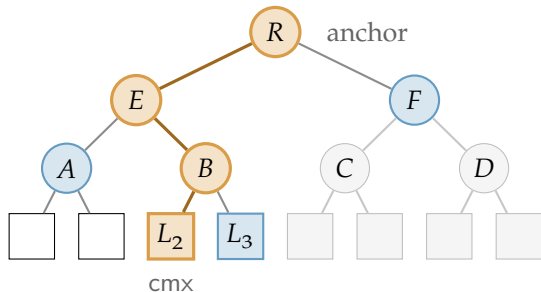
A sealed envelope that can be added to other sealed envelopes.

# Hashes and Merkle trees

$$B = H(L_2, L_3), \quad E = H(A, B), \\ R = H(E, F) = \text{anchor}$$

leaves and nodes in  $\mathbb{F}_p$ .

- Nobody can find two inputs with one output.
- Position  $010_2$ ; siblings  $(L_3, A, F)$ . Depth 32 in the protocol.
- The tree is public; the leaf and the path stay hidden.
- Append only: old roots stay valid.



# A sigma protocol: Schnorr identification

Prover knows  $x$  with  $X = [x] G$ .

**Prover**

**Verifier**

$R = [k] G, k \text{ random}$

$c \text{ random}$

$s = k + c x$

Verifier checks  $[s] G \stackrel{?}{=} R + [c] X$

$x, k, c, s \in \mathbb{F}_q; G, X, R \in \mathcal{E}$ .

## Why it proves, and why it hides

**Special soundness.** Two accepting transcripts with the same  $R$ :

$$(R, c, s), (R, c', s') \implies x = \frac{s - s'}{c - c'}$$

whoever answers two challenges knows  $x$ .

**Zero knowledge.** A simulator with no  $x$  picks  $c$  and  $s$  first:

$$R := [s] G - [c] X$$

the transcript has exactly the real distribution. The verifier learns nothing it could not have made itself.

## Fiat–Shamir: the verifier becomes a hash

$$c := H(X, R, m)$$

- The prover cannot pick  $R$  after  $c$ :  $c$  depends on  $R$ .
- The pair  $(R, s)$  is now a signature on  $m$  — Schnorr.
- **sigma protocol + Fiat–Shamir = Schnorr signature.**
- Zcash's RedPallas is Schnorr on Pallas, with one addition: the key can be re-randomised (later).

# Zero-knowledge proofs, in general

A relation  $\mathcal{R}(x, w)$ : public statement  $x$ , private witness  $w$ .

$$\text{Prove}(x, w) \rightarrow \pi$$

$$\text{Verify}(x, \pi) \in \{0, 1\}$$

- **Complete:** an honest  $\pi$  verifies.
- **Knowledge sound:** from any convincing prover an extractor pulls a  $w$  with  $\mathcal{R}(x, w)$ .
- **Zero knowledge:** a simulator given only  $x$  reproduces  $\pi$ .
- **SNARK:**  $\pi$  short (logarithmic) and non-interactive (Fiat–Shamir).

Schnorr is the proof for  $\mathcal{R} = \{(X, x) : X = [x] G\}$ .

## A polynomial cannot lie everywhere

$$D \neq 0, \deg D \leq d \implies \Pr_{z \leftarrow \mathbb{F}}[D(z) = 0] \leq \frac{d}{|\mathbb{F}|}$$

$D \in \mathbb{F}[X]$  a polynomial over a field  $\mathbb{F}$ ;  $z \in \mathbb{F}$ .

A nonzero polynomial has at most  $d$  roots. Fix  $D$  first, draw  $z$  second: one random probe catches it.

With  $|\mathbb{F}| \approx 2^{254}$  and  $d \approx 2^{14}$  the odds of escape are about  $2^{-240}$ .

# The core of Orchard

## A note

$$n = (d, \text{pk}_d, v, \rho, \psi, \text{rcm})$$

$$m = g_d^* \parallel \text{pk}_d^* \parallel \text{LE}_{64}(v) \parallel \text{LE}_{255}(\rho) \parallel \text{LE}_{255}(\psi),$$
$$\text{cm} = \text{NoteCommit}_{\text{rcm}}(m), \quad \text{cmx} = \text{Extract}(\text{cm}) \in \mathbb{F}_p$$

$$d \in \{0, 1\}^{88}; \text{pk}_d, g_d, \text{cm} \in \mathcal{E}; v \in [0, 2^{64}); \rho, \psi \in \mathbb{F}_p; \text{rcm} \in \mathbb{F}_q; \text{cmx} \in \mathbb{F}_p.$$

$(d, \text{pk}_d)$  the recipient's diversified address

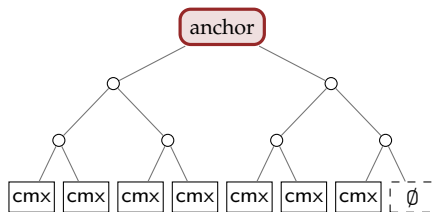
$v$  the value,  $0 \leq v < 2^{64}$

$\rho, \psi$  uniqueness seed and nullifier salt

$\text{rcm}$  the commitment trapdoor

On chain: cmx and a ciphertext. Nothing else about the note.

# The chain's shielded state



- **The commitment tree:** depth 32, append only; every block's final root is an *anchor*.
- **The nullifier set:** the spent half, growing; a nullifier names no leaf.
- A spend proves membership under some anchor and reveals a nullifier.

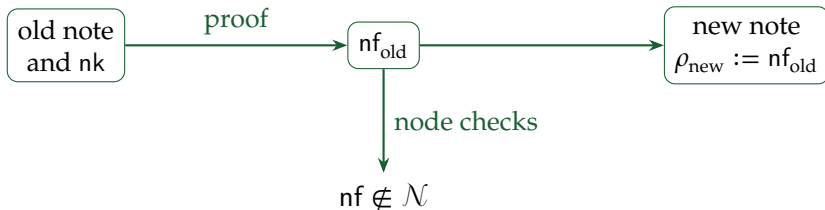
# The nullifier

$$\text{nf} := \text{Extract}([F_{\text{nk}}(\rho) + \psi] G^{\text{nf}} + \text{cm})$$

$$\text{nk}, \rho, \psi, F_{\text{nk}}(\rho) \in \mathbb{F}_p; G^{\text{nf}}, \text{cm} \in \mathcal{E}; \text{nf} \in \mathbb{F}_p.$$

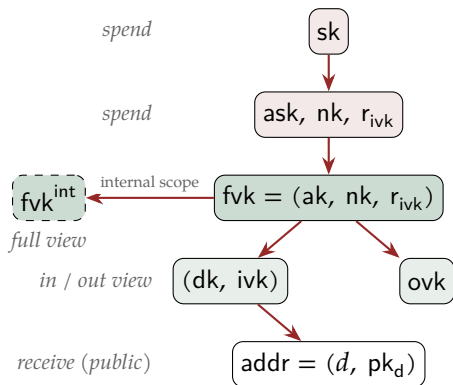
- Keyed by  $\text{nk}$ , the owner's nullifier key;  $F_{\text{nk}}$  is Poseidon, a hash the circuit evaluates natively.
- Seeded by  $\rho$ , salted by  $\psi$ , bound to  $\text{cm}$ .
- Chaining  $\rho_{\text{new}} := \text{nf}_{\text{old}}$ : every note inherits the nullifier of the note spent beside it, so  $\rho$  is unique.
- The sum is taken in  $\mathbb{F}_p$ , then read as a scalar ( $p < q$ ).
- Unique, and unlinkable to  $\text{cm}$  without  $\text{nk}$ .

## The proof and the node check different things



The set  $\mathcal{N}$  holds every spent nullifier. Reject repeats; insert only when the transaction is accepted.

# Keys



$ask, nk, r_{ivk} := \text{PRFexpand}_{sk}(06), (07), (08)$

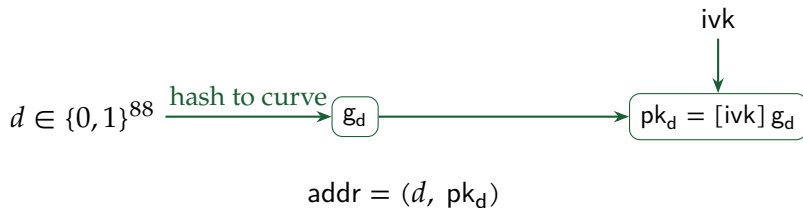
$ak := [ask] G^{\text{SpendAuth}}$

$ivk := \text{Commit}_{r_{ivk}}^{ivk}(\text{Extract}(ak), nk)$   
 $(dk, ovk) := \text{PRFexpand}_{r_{ivk}}(82 \parallel \text{Extract}(ak) \parallel nk)$

$d := \text{FF1}_{dk}(j), g_d := \text{GroupHash}(d),$   
 $pk_d := [ivk] g_d$

$ask, r_{ivk} \in \mathbb{F}_q; nk \in \mathbb{F}_p; ak \in \mathcal{E}; ivk \in \mathbb{F}_p$ , used as a scalar ( $p < q$ );  $dk, ovk$  byte strings;  
 $d \in \{0, 1\}^{88}; pk_d \in \mathcal{E}.$

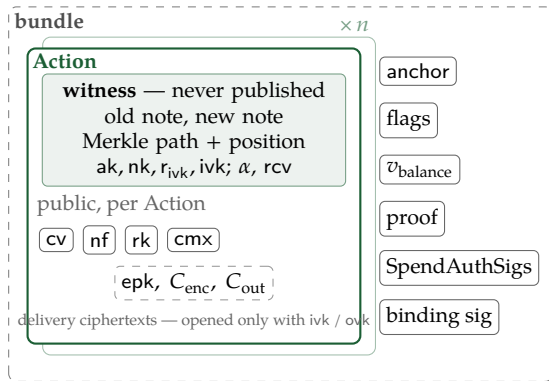
# Diversified addresses



$d \in \{0, 1\}^{88}; g_d, pk_d \in \mathcal{E}; ivk \in \mathbb{F}_p$  as a scalar.

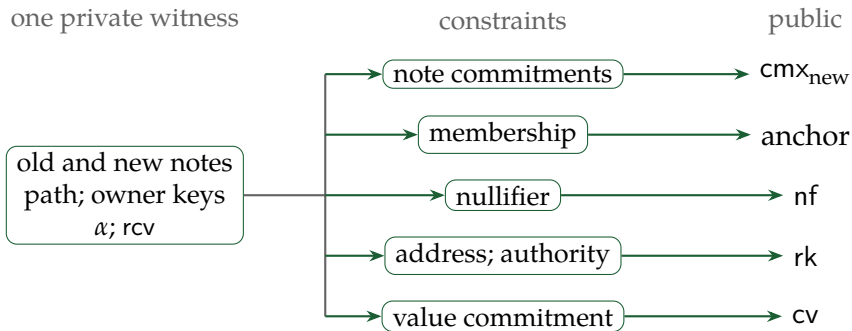
- Some  $2^{88}$  addresses per key, all paid into one wallet.
- Unlinkable to each other, even by the payer: two addresses share  $ivk$  only as a discrete log.
- One  $ivk$  detects every payment to every address.

# The Action



- One Action spends one note and creates one note.
- A dummy note pads either side; every Action looks alike.
- One proof per bundle covers all its Actions.

# The statement, not the secret



Every constraint uses the same notes and the same owner keys.

## The statement

| check       | clause | the proof establishes                                           |
|-------------|--------|-----------------------------------------------------------------|
| exists      | A3     | a Merkle path from $cm_{old}$ to the anchor                     |
| not twice   | A5     | the nullifier $nf$ is this note's serial, keyed by $nk$         |
| conserved   | A4     | the commitment $cv$ hides $v_{old} - v_{new}$ , both $< 2^{64}$ |
| authorised  | A6, A7 | the key $rk$ disguises $ak$ ; $ak, nk$ sit behind the address   |
| well-formed | A1, A2 | both commitments open to notes; $\rho_{new} = nf_{old}$         |
| gated       | A8–A10 | spends, outputs, cross-address payments switched by flags       |

The witness: both notes, the path and position, the keys,  $\alpha$ ,  $rcv$ .

The checks the proof cannot make: is the anchor an earlier block's, is  $nf$  new.

## Conservation is a cancellation

$$cv_i = [v_{\text{old},i} - v_{\text{new},i}] V + [rcv_i] R \quad \text{per Action}$$

Let  $b$  be the public net value leaving the pool ( $v_{\text{balance}}$ ). The public binding key:

$$bvk := \sum_i cv_i - [b] V = \left[ \underbrace{\sum_i (v_{\text{old},i} - v_{\text{new},i})}_{\Delta: \text{ imbalance}} - b \right] V + \left[ \sum_i rcv_i \right] R$$

$v \in [0, 2^{64})$ ;  $rcv_i \in \mathbb{F}_q$ ;  $cv_i, bvk, V, R \in \mathcal{E}$ ;  $b$  a signed 64-bit integer;  $\Delta \in \mathbb{F}_q$ .

For a fully shielded transfer,  $b$  is the fee.

## Why the binding signature enforces balance

A signature under  $\text{bvk}$  on base  $R$  proves knowledge of  $k$  with  $\text{bvk} = [k] R$ . With  $s := \sum_i \text{rcv}_i$ :

$$[\Delta] V = [k - s] R$$

$$k, s, \Delta, \lambda \in \mathbb{F}_q.$$

- Balanced:  $\Delta = 0$ , and the author knows  $k = s$ .
- Unbalanced:  $\Delta \neq 0$  gives  $\lambda = (k - s) \Delta^{-1}$  with  $V = [\lambda] R$ , the forbidden relation.

Schnorr again: a proof of knowledge of a discrete log, of a key that exists only when the books balance.

## Field equality is not yet integer equality

$$q - 1 = -1, \quad q = 0 \quad \text{in } \mathbb{F}_q$$

So the proof also constrains every amount:

$$0 \leq v < 2^{64}$$

Bounded amounts and a bounded number of Actions keep an integer imbalance from wrapping around modulo  $q$ .

## Authorisation without a fixed key

$$ak = [ask] G^{\text{SpendAuth}}, \quad pk_d = [ivk] g_d,$$

$$ivk = \text{Commit}_{r_{ivk}}^{ivk}(\text{Extract}(ak), nk)$$

$$rk := ak + [\alpha] G^{\text{SpendAuth}} = [rsk] G^{\text{SpendAuth}}, \quad rsk := ask + \alpha$$

$ask, \alpha, rsk \in \mathbb{F}_q$ ;  $ak, rk \in \mathcal{E}$ ;  $ivk, nk \in \mathbb{F}_p$ .

- One  $ak$  per wallet; publishing it would stamp every spend.
- Fresh  $\alpha$  per spend:  $rk$  is a uniform point, unlinkable.
- The signature is Schnorr under  $rk$ ; the circuit proves  $rk$  was formed from the  $ak$  behind the spent note's address.

# Delivery

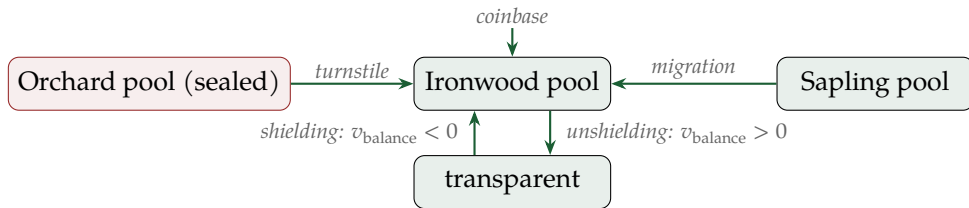
$$\text{epk} := [\text{esk}] g_d, \quad s := [\text{esk}] \text{pk}_d = [\text{ivk}] \text{epk}, \quad K := \text{KDF}(s, \text{epk})$$

$$C_{\text{enc}} := \text{AEAD}_K(0x03, d, v, \text{rseed}, \text{memo})$$

$\text{esk} \in \mathbb{F}_q; g_d, \text{pk}_d, \text{epk}, s \in \mathcal{E}; K \in \{0, 1\}^{256}; C_{\text{enc}}: 580 \text{ bytes}.$

- One-shot Diffie–Hellman to the diversified base, then ChaCha20-Poly1305.
- No address on chain: the recipient trial-decrypts every output with ivk. A wrong key is rejected by the tag.
- The ciphertext  $C_{\text{out}}$  holds  $(\text{pk}_d, \text{esk})$  under a key from ovk.

# The doors



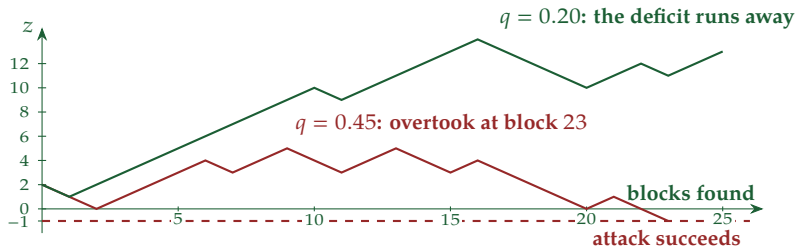
Every crossing is public in amount. Inside a pool, nothing is. Each pool's total is public and may never go negative.

## What consensus adds

- A nullifier must not repeat within a transaction or across the valid chain.
- Transactions apply in block order: check  $nf \notin \text{set}$ , insert  $nf$ , append  $\text{cmx}$ .
- An anchor must be the final root of an earlier block — no spending an unmined note.
- A reorganisation rolls back tree, nullifier set and pool totals together.

The same shape as the UTXO set: an ordered, stateful check — over serial numbers that name no output.

# The catch-up probability



Honest lead  $z$ ; each block moves it  $+1$  with probability  $p$ ,  $-1$  with probability  $q$ .

$$a_z = \Pr[\text{lead ever reaches } -1] = (q/p)^{z+1} \quad (q < p)$$

here  $p, q \in [0, 1]$  are hashrate shares,  $p + q = 1$ , not the primes.

Each block of deficit multiplies the attacker's chance by  $q/p$ .

**How Halo 2 proves**

## The claim becomes a table

"I know  $x$  with  $x^3 + x + 5 = 35$ " over  $\mathbb{F}_{97}$ ; the witness  $x = 3$ :

$$3 \xrightarrow{\text{square}} 9 \xrightarrow{\times 3} 27 \xrightarrow{+3} 30 \xrightarrow{+5} 35.$$

One operation per row, one gate equation on every row:

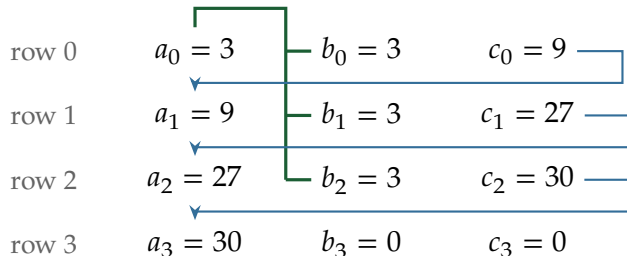
$$q_M a b + q_L a + q_R b + q_O c + q_C + \text{PI} = 0$$

every cell in  $\mathbb{F}_{97}$ .

| row | advice |     |     | fixed |       |       |       |       | instance |                     |
|-----|--------|-----|-----|-------|-------|-------|-------|-------|----------|---------------------|
|     | $a$    | $b$ | $c$ | $q_M$ | $q_L$ | $q_R$ | $q_O$ | $q_C$ | PI       |                     |
| 0   | 3      | 3   | 9   | 1     | 0     | 0     | -1    | 0     | 0        | $x \cdot x = x^2$   |
| 1   | 9      | 3   | 27  | 1     | 0     | 0     | -1    | 0     | 0        | $x^2 \cdot x = x^3$ |
| 2   | 27     | 3   | 30  | 0     | 1     | 1     | -1    | 0     | 0        | $x^3 + x = s$       |
| 3   | 30     | 0   | 0   | 0     | 1     | 0     | 0     | 5     | -35      | $s + 5 = 35$        |

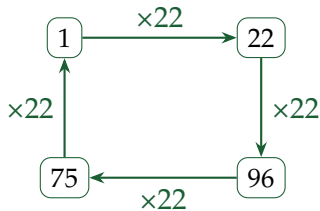
Advice: the prover's secret trace. Fixed: the circuit. Instance: the public 35. Copy constraints wire the rows; range checks are lookups into a table.

## Equal values must be wired together



Equal numbers in a table are not a constraint. These *copy constraints* are enforced separately, by a permutation argument.

## Give the rows multiplicative labels



$$\omega = 22, \quad \omega^2 = 96 = -1, \quad \omega^4 = 1$$

$$H = \{1, 22, 96, 75\} \subset \mathbb{F}_{97}$$

Row  $i$  gets the label  $\omega^i$ : a cycle of field values.

## Rows become polynomials

Row  $i$  lives at  $\omega^i \in H$ . Interpolate each column:

$$a(1) = 3, a(22) = 9, a(96) = 27, a(75) = 30 \implies a(X) = 90 + 61X + 22X^2 + 24X^3$$

$$G(X) := q_M ab + q_L a + q_R b + q_O c + q_C + \text{PI}$$

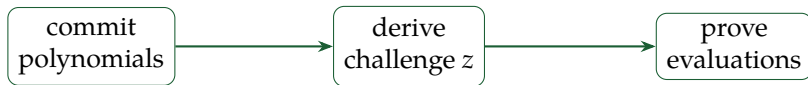
$a, b, c, q_\bullet, \text{PI} \in \mathbb{F}_{97}[X]$  of degree  $< 4$ ;  $G \in \mathbb{F}_{97}[X]$  of degree  $\leq 9$ .

$$Z_H(X) := \prod_{h \in H} (X - h) = X^4 - 1$$

$$\text{every gate holds} \iff G \text{ vanishes on } H \iff Z_H \mid G \iff G = t \cdot Z_H$$

A correct computation *is* a polynomial identity. A wrong one cannot be: the lie  $x^2 = 10$  in row 0 leaves a remainder  $24(1 + X + X^2 + X^3) \neq 0$ .

## One random point



1. The prover *commits* to  $a, b, c$  and  $t$ .
2. The challenge  $z := H(\text{transcript})$  — Fiat–Shamir.
3. The prover opens  $a(z), b(z), c(z), t(z)$ .
4. The verifier checks one equation:

$$G(z) \stackrel{?}{=} t(z) Z_H(z)$$

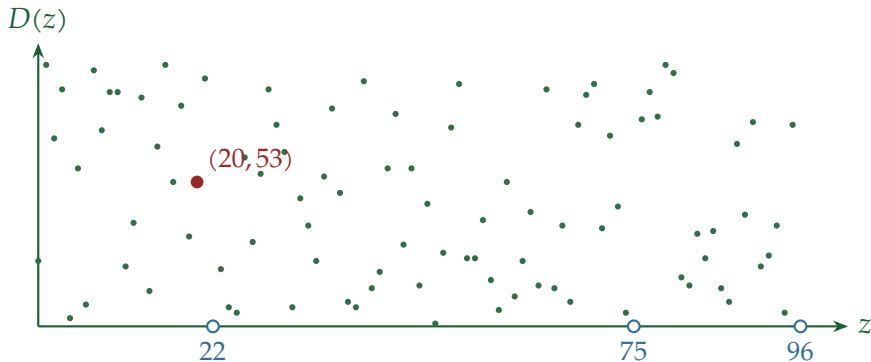
$z \in \mathbb{F}_{97}$  in the toy;  $z \in \mathbb{F}_p$  in the protocol.

Honest,  $z = 20$ :  $G(20) = 75 = 67 \cdot 46$ . The row-0 lie:  $G^*(20) = 20 \neq 64$ .

A false identity is frozen before  $z$  is drawn; it survives only if  $z$  is one of its few roots.

## A nonzero polynomial has few roots

The remainder  $D(X) = 24(1 + X + X^2 + X^3)$  over  $\mathbb{F}_{97}$ : zeros at 22, 75, 96 only.



For uniform  $z$  this cheat passes with probability  $3/97$ ; over the protocol's field, about  $2^{-240}$ .

# Committing to a polynomial

$$P := \langle \vec{a}, \vec{G} \rangle + [r] H, \quad \vec{a} = (a_0, \dots, a_{d-1}), \quad \vec{G} \text{ independent points}$$

$$a(z) = \langle \vec{a}, \vec{b} \rangle, \quad \vec{b} := (1, z, z^2, \dots, z^{d-1})$$

$a_i, r, z \in \mathbb{F}_p$ ;  $\vec{G}, H, P \in \mathcal{U}$ , the Vesta curve, whose scalar field is  $\mathbb{F}_p$ .

- A Pedersen commitment to the coefficient vector: hiding, binding.
- Opening at  $z$  is an inner product with a public vector.
- The problem: convince the verifier of  $\langle \vec{a}, \vec{b} \rangle = v$  in fewer than  $d$  group elements, without sending  $\vec{a}$ .

# The inner-product argument: one fold

Split every vector in half and publish two cross terms:

$$\begin{aligned} L &:= \langle \vec{a}_{\text{lo}}, \vec{G}_{\text{hi}} \rangle + [\ell] H + [\langle \vec{a}_{\text{lo}}, \vec{b}_{\text{hi}} \rangle] U, \\ R &:= \langle \vec{a}_{\text{hi}}, \vec{G}_{\text{lo}} \rangle + [r_R] H + [\langle \vec{a}_{\text{hi}}, \vec{b}_{\text{lo}} \rangle] U. \end{aligned}$$

Challenge  $u$ ; both sides fold:

$$\vec{a}' := u \vec{a}_{\text{lo}} + u^{-1} \vec{a}_{\text{hi}}, \quad \vec{G}' := u^{-1} \vec{G}_{\text{lo}} + u \vec{G}_{\text{hi}}, \quad \vec{b}' := u^{-1} \vec{b}_{\text{lo}} + u \vec{b}_{\text{hi}}$$

$$P + [v] U + [u^2] L + [u^{-2}] R = \langle \vec{a}', \vec{G}' \rangle + [r + u^2 \ell + u^{-2} r_R] H + [\langle \vec{a}', \vec{b}' \rangle] U$$

$$u \in \mathbb{F}_p^\times; L, R, U, H \in \mathcal{U}; \ell, r_R \in \mathbb{F}_p.$$

The same claim at half the length. The verifier folds  $\vec{G}$  and  $\vec{b}$  itself and never sees  $\vec{a}$ .

## The recursion

After  $k = \log_2 d$  rounds every vector has length 1. The proof:  $2k$  points  $(L_j, R_j)$ , one scalar  $a$ , one blinder  $r^*$ .

$$P + [v] U + \sum_{j=1}^k ([u_j^2] L_j + [u_j^{-2}] R_j) \stackrel{?}{=} [a] G^{(0)} + [r^*] H + [a \cdot b^{(0)}] U$$

$a, b^{(0)}, r^*, u_j \in \mathbb{F}_p; G^{(0)}, L_j, R_j \in \mathcal{U}$ .

- The scalar  $b^{(0)} = \prod_j (u_j^{-1} + u_j z^{2^{j-1}})$ , with  $u_1$  the last challenge: logarithmic work.
- The point  $G^{(0)} = \langle \vec{s}, \vec{G} \rangle$ : one multi-scalar multiplication, shared across a batch of proofs.
- Binding: two openings would give a discrete-log relation among  $\vec{G}, H$ ; an extractor rewinds three challenges per round.

## Binding does not by itself give privacy

An unblinded evaluation is an equation in the secret coefficients:

$$f(z) = f_0 + f_1z + \cdots + f_{d-1}z^{d-1}.$$

random unused rows  
mask the evaluations

random commitment blinders  
hide the coefficients

Zero knowledge: the proof reveals nothing beyond the public statement.

# The Orchard circuit

- Size:  $2^{11}$  rows, 10 advice columns; arithmetic over  $\mathbb{F}_p$  so Pallas points  $cm, rk, cv$  are native; commitments on Vesta.
- Gadgets: Sinsemilla (commitments, tree), Poseidon (nullifier), elliptic-curve arithmetic, lookups for range checks.
- The instance:  $anchor, cv, nf, rk, cmx, flags$ .
- Generators from a hash: no trusted setup, no ceremony.
- Proof size  $2720 + 2272a$  bytes for  $a$  Actions: 4992 for one.

Every clause of the statement is a set of rows; one proof, one random point, one inner-product argument.

## Back to the transaction

|                   |                                        |
|-------------------|----------------------------------------|
| Proof             | the hidden notes satisfy the statement |
| Spend signatures  | the owners authorise this transaction  |
| Binding signature | the committed amounts balance          |
| Node state        | anchor allowed; nullifiers unused      |

No single equation replaces the other checks.

## The whole picture

| check      | Bitcoin           | Ironwood                           |
|------------|-------------------|------------------------------------|
| exists     | UTXO lookup       | Merkle path, inside the proof      |
| not twice  | delete the entry  | nullifier, checked against the set |
| conserved  | add the amounts   | $\sum cv$ , binding signature      |
| authorised | script, signature | A6/A7, signature under rk          |

The validator learns that the four checks pass — and nothing about the note.